Executive Summary

PowerCo SME Churn Analysis

Recommended Solution & Business Impact

- Deploy a Targeted Retention Strategy: Launch proactive retention campaigns offering a 20% discount exclusively to the top 28% highest-risk SME customers identified by our Random Forest model.

Situation

- PowerCo faces increasing competition in the SME energy market, experiencing a baseline annual customer churn rate of ~9.7%.
- Acquiring new SME energy clients is significantly more costly than retaining existing ones, making customer retention vital to long-term profitability.

Complication

- Price Sensitivity is Minor Driver: Analysis reveals price changes account for only a small fraction of overall churn. Mass discounting across all customers would heavily erode margins without solving churn.
- Severe Class Imbalance: In our PowerCo customer churn dataset, out of thousands of SME clients, only a tiny fraction (around 10%, or 366 out of 3,652 clients in our test set) actually chose to leave (churn), while roughly 90%stayed. This creates a heavily skewed or "imbalanced" environment where the non-churners completely drown out the churners.

Key Findings

- Model Performance: Calibrated Random Forest model achieves 86% overall accuracy and captures 28% of all churners (5x improvement over baseline) using optimized probability thresholds.